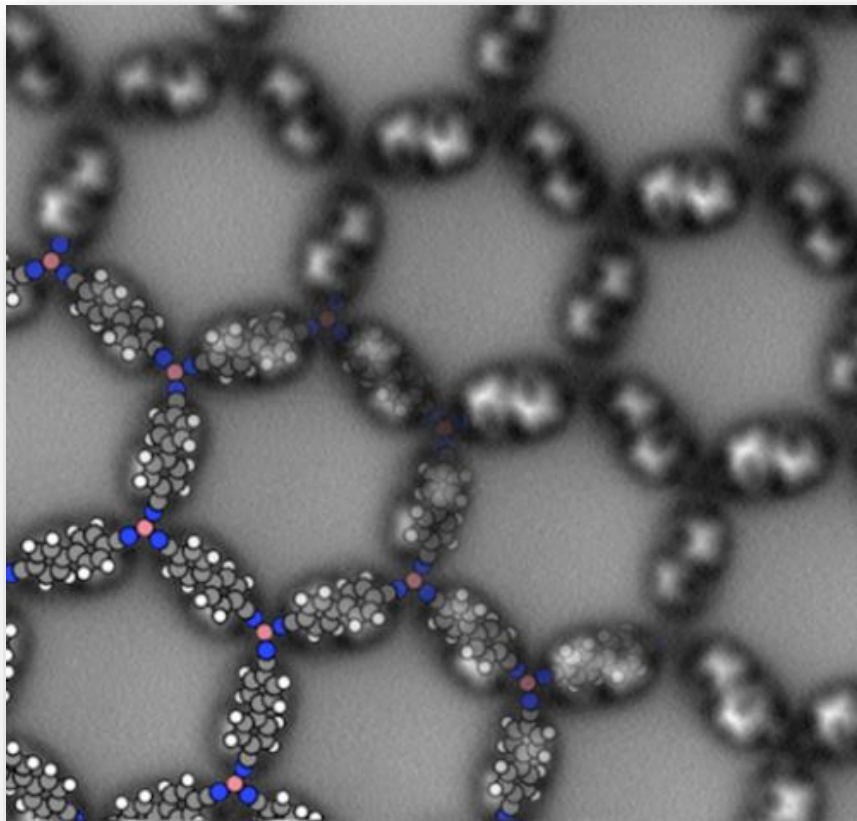




2D METAL ORGANIC FRAMEWORKS

Samantha Bacchi

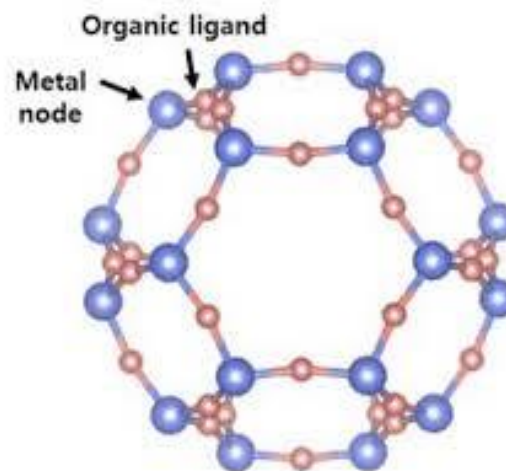
sbacch2@uic.edu

Department of Chemical Engineering

May 7th, 2020

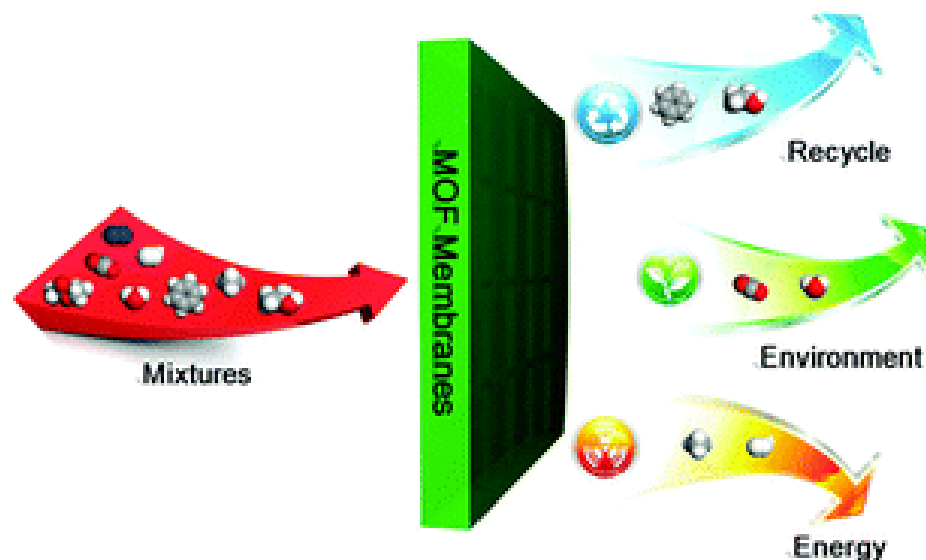
Introduction

- Metal organic frameworks (MOFs) are composed of inorganic metallic nodes with organic linkers, or ligands, held together through intermolecular forces.
- Research attempts include obtaining both macro and nano-crystalline MOFs of various shapes and sizes by altering composition through linker design and post-synthetic modification.
- The desired functionality of a MOF can be obtained by selecting the appropriate organic ligand/metallic node combination.
- The most conductive MOFs are those with a planar, graphene-like structure where the charge carriers are confined to the 2D lattice.

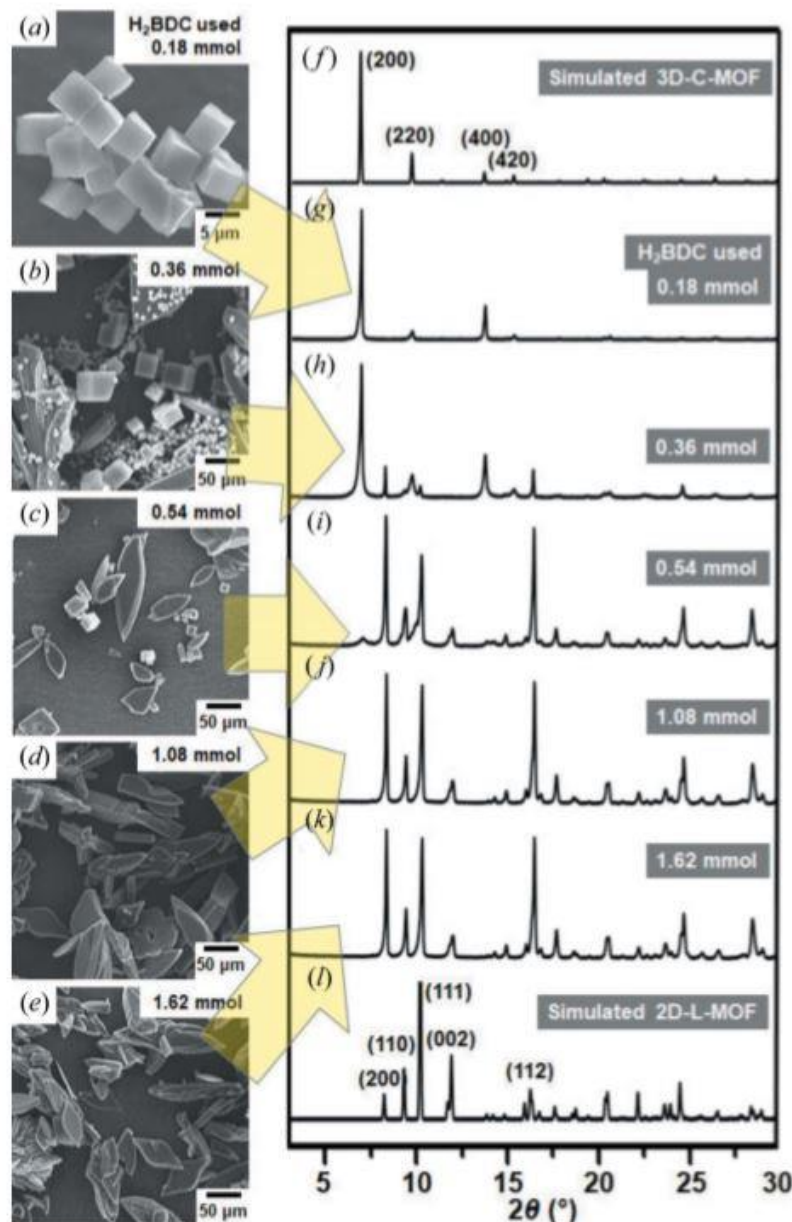


Importance

- The goal of current research is to develop MOFs which have inherently high electronic properties which is not diminished the intended substrate.
- Importance of MOFs are in healthcare, environmental, and technological applications such as gas separations, energy storage and conversion, solar and fuel cells, capacitors and catalysis.



Methods & Results



- MOF synthesis under solvothermal conditions is the most common method:
 - Metal salt hydrate complexes are combined with the organic linkers and solvents.
 - Mixture is placed into an autoclave and into an oven for 24+ hours.
 - Crystalline product is then obtained by vacuum filtration and rinsed. Once dry,
 - Analyzed using x-ray diffraction (XRD) and a scanning electron microscope (SEM).
- XRD and SEM confirm the MOFs synthesized are comprised of desired structures.
- Electrochemical properties are determined using a standard three-electrode test pool and electrochemical impedance spectroscopy.

Challenges

- Major change is realistically obtaining the designed MOF with the desired topology and functionality.
- Synthesizing crystals which solely grow along the vertical direction without affecting the two other lateral directions.
- MOFs inherently have low electrical conductivity which must be enhanced through better design to increase the concentration and mobility of charge carriers.
- Strong interactions between the 2D MOFs and their metal substrates hinder the electronic properties of the MOF alone.

Future Opportunities

- Future work should focus on the development of MOFs with higher conductivity.
 - Increasing charge carriers by thermal or photo-excitation, doping, and hole or electron injection.
 - Increasing charge mobility by minimizing grain boundaries and reducing the number of charge trapping sites
- Post-synthetic modification incorporates new molecules to the MOFs that act as bridges between neighboring MOF units which promote charge transport.
- MOFs development should limit doping and instead look to redox active organic linkers which have little affect on structure and substantially increase electrical conductivity.

References

- Day, R. W., Bediako, D. K., Rezaee, M., Parent, L. R., Skorupskii, G., Arguilla, M. Q., . . . Dinca, M. (2019). Single Crystals of Electrically Conductive Two-Dimensional Metal-Organic Frameworks: Structural and Electrical Transport Properties. *ACS Central Science*(5), 1959-1964.
- Dhakshinamoorthy, A., Asiri, A. M., & Garcia, H. (2019). 2D Metal-Organic Frameworks as Multifunctional Materials in Heterogeneous Catalysis and Electro/Photocatalysis. *Advanced Materials*(31).
- Duan, J., Li, Y., Pan, Y., Behera, N., & Jin, W. (2019). Metal-Organic Framework Nanosheets: An Emerging Family of Multifunctional 2D Materials. *Journal of Coordination Chemistry Reviews*(395), 25-45.
- Ko, M., Mendecki, L., & Mirica, K. A. (2018). Conductive Two-Dimensional Metal-Organic Frameworks as Multifunctional Materials. *ChemComm*(54), 7873-7891.
- Liu, X., Valentine, H. L., Pan, W.-P., Cao, Y., & Yan, B. (2016). 2D Metal-Organic Frameworks: Syntheses, Structures, and Electrochemical Properties. *Journal of Inorganica Chimica Acta*(447), 162-167.